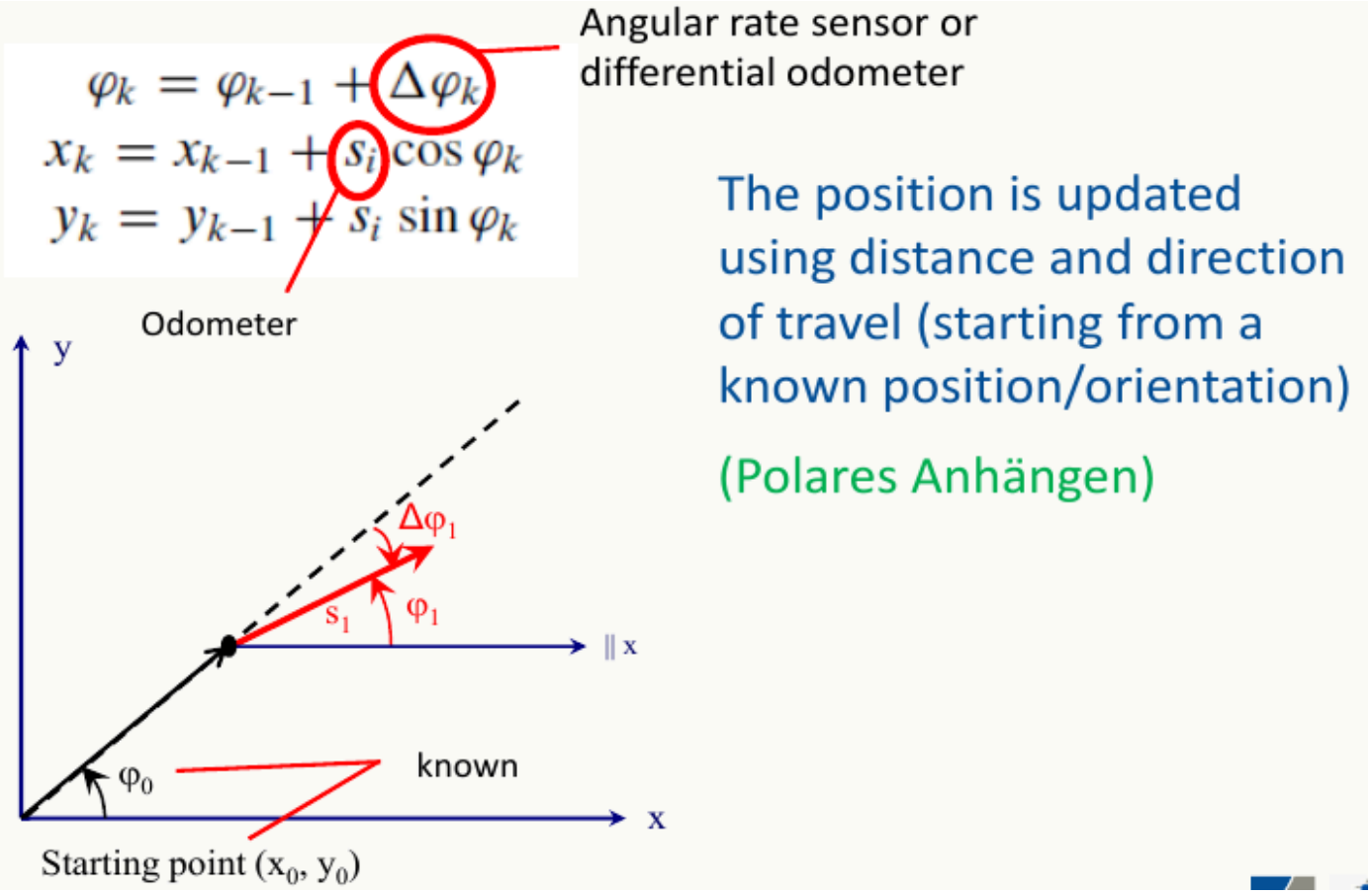
Ms04: Inertial Navigation

Gyroscope	Accelerometer	Odometer	Differential odometer
Change in angle	Change in velocity	Travelled distance	Change in Direction

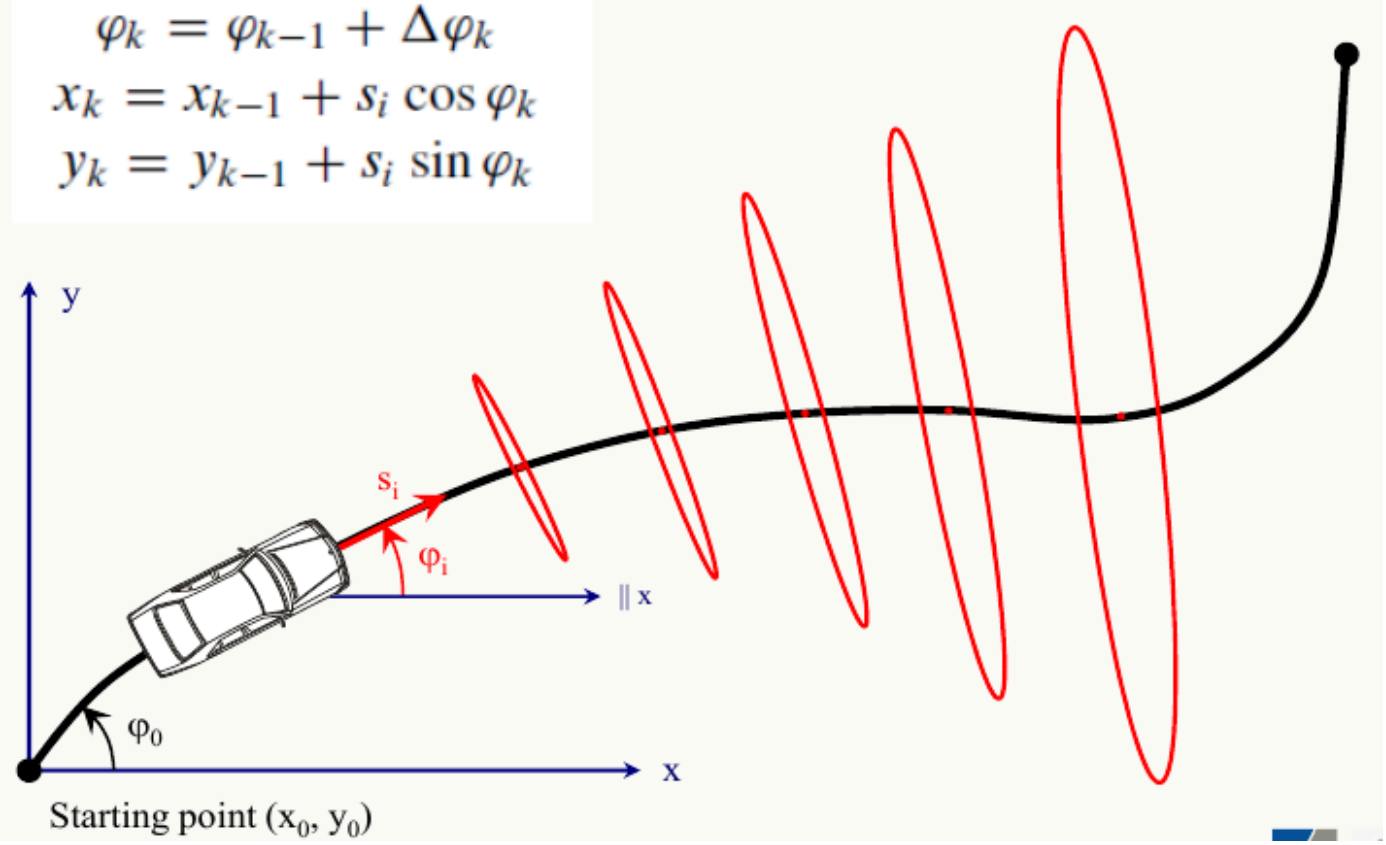
- 2D: Dead Reckoning
- 3D: Inertial Navigation

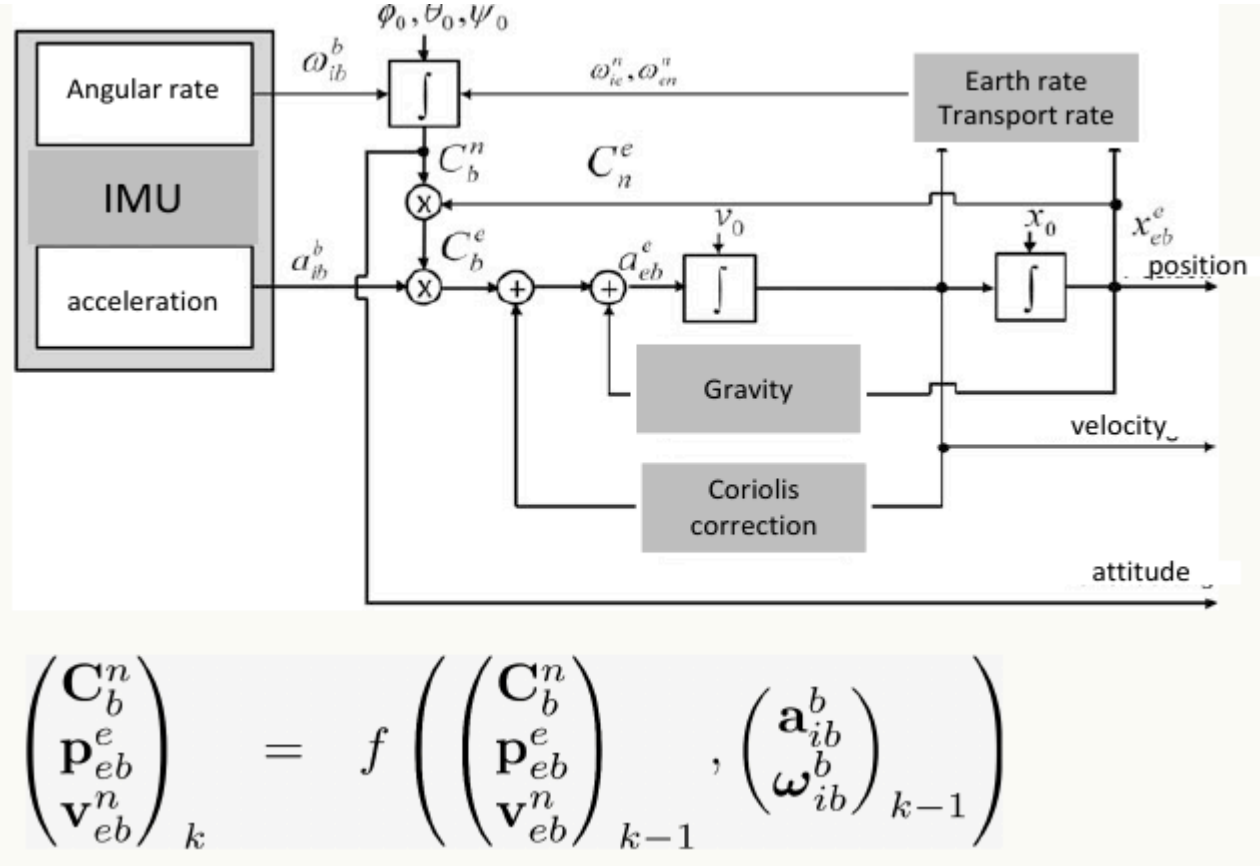
Dead Reckoning



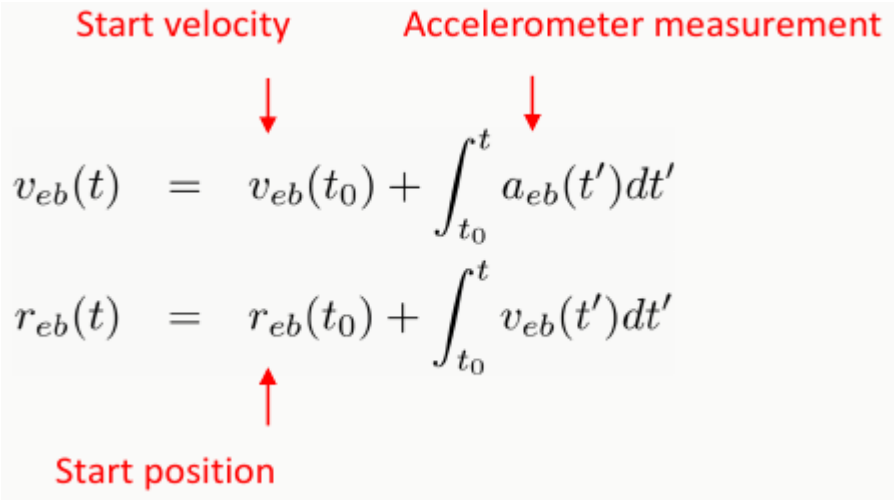
- **Direction**:
  - Compass
  - Angle to known landmarks
  - Angle to stars, celestial objects
  - Gyroscope (angular rate)
  - Differential odometer
  - **Angle** or integrated **angular rate** measurements
- **Speed**:

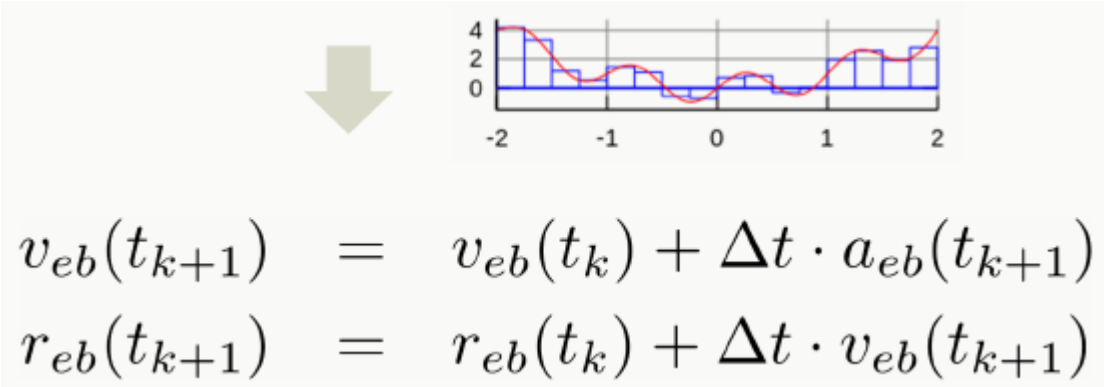
- Counting steps (pedestrian navigation)
- Counting knots on a rope (Ship)
- Odometer
- Accelerometer
- **Distance travelled** per time or **velocity** over ground measurements





1D/2D Inertial Navigation

- e: earth fixed system
- b: body system
  - constraint to move on straight line pependicular to gravity -> no orientation
- accelerometer measure along the direction of motion, for now ignoring any motion of the Earth
- Numerical Integration
  - Midpoint or rectangular rule
    - $\int_a^b f(x)dx \approx (b-a)f(\frac{a+b}{2})$
  - Trapezoidal rule
    - $\int_a^b f(x)dx \approx (b-a)f(\frac{f(y)+f(b)}{2})$
  - More complicated and accurate integration schemes
    - e.g. Simpsons-Rule
- 

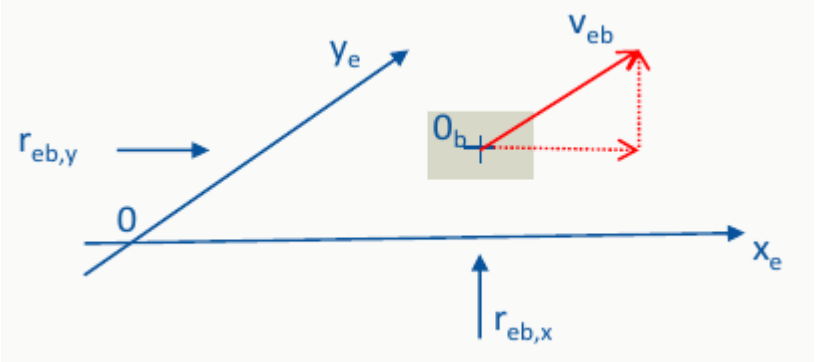
- -> With **Numerical Integration**
- 

2D Example

- Motion constraint to a plane perpendicular to gravity
- Orientation possible to any direction within the plan
- One angular and two linear degrees of freedom

- 2 accelerometers (sensitive axis horizontally along the body axis)

$$\begin{bmatrix} v^e_{eb,x} \\ v^e_{eb,y} \end{bmatrix}_{k+1} = \begin{bmatrix} v^e_{eb,x} \\ v^e_{eb,y} \end{bmatrix}_{k} + \Delta t \cdot \begin{bmatrix} a^e_{eb,x} \\ a^e_{eb,y} \end{bmatrix}_{k+1}$$
$$\begin{bmatrix} r^e_{eb,x} \\ r^e_{eb,y} \end{bmatrix}_{k+1} = \begin{bmatrix} r^e_{eb,x} \\ r^e_{eb,y} \end{bmatrix}_{k} + \Delta t \cdot \begin{bmatrix} v^e_{eb,x} \\ v^e_{eb,y} \end{bmatrix}_{k+1}$$

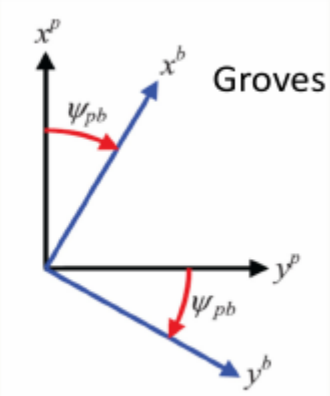
- 

- acceleration $a^e_{eb,x}$ and $a^e_{eb,y}$ along the e-axis
  - **not** what the sensor measures:
  - sensor reading in b-frame
  - $a^b_{eb,x}$ and $a^b_{eb,y}$
- Transformation of measurement (b-frame) to the e-frame:

$$\begin{pmatrix} a^e_{eb,x} \\ a^e_{eb,y} \end{pmatrix}_k = C^e_{b,k} \begin{pmatrix} a^b_{eb,x} \\ a^b_{eb,y} \end{pmatrix}_k$$

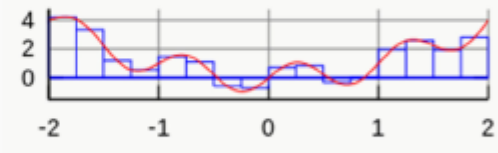
$$C^e_{b,k} = \begin{pmatrix} \cos \Psi_{eb,k} & -\sin \Psi_{eb,k} \\ \sin \Psi_{eb,k} & \cos \Psi_{eb,k} \end{pmatrix}$$

angle between b and e-frame



- $\Psi_{eb,k}$ via Integration of angular rates
  - $\omega^b_{eb}(t')$ via Gyroscope measurements:

- $$\Psi_{eb}(t) = \Psi_{eb}(t_0) + \int_{t_0}^{t} \omega_{eb}^{b}(t')dt'$$

  $$\Psi_{eb,k} = \Psi_{eb,k-1} + \Delta t \cdot \omega_{eb,k}^{b}$$

Inertial Navigation - 2D - No Gravity

1. Attitude update $\Psi_{eb,k} = \Psi_{eb,k-1} + \Delta t \cdot \omega_{eb,k}^{b}$
2. Transformation of accelerometer measurement to the system of interest (**here e**)

$$\begin{pmatrix} a_{eb,x}^{e} \\ a_{eb,y}^{e} \end{pmatrix}_k = \begin{pmatrix} \cos\Psi_{eb,k} & -\sin\Psi_{eb,k} \\ \sin\Psi_{eb,k} & \cos\Psi_{eb,k} \end{pmatrix} \begin{pmatrix} a_{eb,x}^{b} \\ a_{eb,y}^{b} \end{pmatrix}_k$$

3. velocity update

$$\begin{pmatrix} v_{eb,x}^{e} \\ v_{eb,y}^{e} \end{pmatrix}_k = \begin{pmatrix} v_{eb,x}^{e} \\ v_{eb,y}^{e} \end{pmatrix}_{k-1} + \Delta t \cdot \begin{pmatrix} a_{eb,x}^{e} \\ a_{eb,y}^{e} \end{pmatrix}_k$$

4. Position update

$$\begin{pmatrix} r_{eb,x}^{e} \\ r_{eb,y}^{e} \end{pmatrix}_k = \begin{pmatrix} r_{eb,x}^{e} \\ r_{eb,y}^{e} \end{pmatrix}_{k-1} + \Delta t \cdot \begin{pmatrix} v_{eb,x}^{e} \\ v_{eb,y}^{e} \end{pmatrix}_k$$

Dead reckoning

- From 2D to 3D:
  - Attitude update **needs integration of 3 angular** instead of 1
    - -> Integration of angular rates in 3D: NOT by integrating Euler angles independently
  - Gravitation not measured by accelerometer
    - Needs to be considered when modeling velocity
    - **Local gravity level** and **current orientation** with respect to gravity needs to be known

Integration of Angular Rates in 3D

Small angle approximation for small angles γ

- $\sin\gamma = \gamma$
- $\cos\gamma = 1$
- $\gamma \cdot \gamma = 0$
- $$\begin{aligned} \mathbf{C}^{\alpha}_{\beta} &= \begin{bmatrix} \cos\theta\cos\psi & -\cos\phi\sin\psi + \sin\phi\sin\theta\cos\psi & \sin\phi\sin\psi + \cos\phi\sin\theta\cos\psi \\ \cos\theta\sin\psi & \cos\phi\cos\psi + \sin\phi\sin\theta\sin\psi & -\sin\phi\cos\psi + \cos\phi\sin\theta\sin\psi \\ -\sin\theta & \sin\phi\cos\theta & \cos\phi\cos\theta \end{bmatrix} \\ &\approx \begin{bmatrix} 1 & -\psi_{\alpha\beta} & \theta_{\alpha\beta} \\ \psi_{\alpha\beta} & 1 & -\phi_{\alpha\beta} \\ -\theta_{\alpha\beta} & \phi_{\alpha\beta} & 1 \end{bmatrix} = \mathbb{1}_3 - \delta\mathbf{\Psi}_{\alpha\beta} = \mathbb{1}_3 + \delta\mathbf{\Psi}_{\alpha\beta} \end{aligned}$$
  $$\delta\mathbf{\Psi} = \begin{bmatrix} 0 & -\delta\psi & \delta\theta \\ \delta\psi & 0 & -\delta\phi \\ -\delta\theta & \delta\phi & 0 \end{bmatrix}$$
- $$\lim_{\delta t \to 0} \begin{bmatrix} 0 & -\frac{\delta\psi}{\delta t} & \frac{\delta\theta}{\delta t} \\ \frac{\delta\psi}{\delta t} & 0 & -\frac{\delta\phi}{\delta t} \\ -\frac{\delta\theta}{\delta t} & \frac{\delta\phi}{\delta t} & 0 \end{bmatrix}$$
  Angular rates
  $$\mathbf{\Omega}_{\alpha\beta} = \begin{pmatrix} 0 & -\omega_{\alpha\beta,z} & \omega_{\alpha\beta,y} \\ \omega_{\alpha\beta,z} & 0 & -\omega_{\alpha\beta,x} \\ -\omega_{\alpha\beta,y} & \omega_{\alpha\beta,x} & 0 \end{pmatrix}$$
- ns
  $\rightarrow$

Differential equation for rotations/orientations

$$\dot{\boldsymbol{C}}^{\alpha}_{\beta} = \boldsymbol{C}^{\alpha}_{\beta}(t)\mathbf{\Omega}^{\alpha}_{\alpha\beta}(t)$$

$\rightarrow$

Integration between two timestamps

$$\boldsymbol{C}^{\alpha}_{\beta}(t) = \boldsymbol{C}^{\alpha}_{\beta}(0)\exp(\int_0^t \mathbf{\Omega}^{\beta}_{\alpha\beta}(t)dt)$$
$$\boldsymbol{C}^{\alpha}_{\beta}(t+\delta t) = \boldsymbol{C}^{\alpha}_{\beta}(t)\exp(\int_t^{t+\delta t} \mathbf{\Omega}^{\beta}_{\alpha\beta}(t')dt')$$

$$\mathbf{C}^{\alpha}_{\beta}(t+\delta t) = \mathbf{C}^{\alpha}_{\beta}(t)\exp\left(\int_{t}^{t+\delta t}\mathbf{\Omega}^{\beta}_{\alpha\beta}(t')dt'\right)$$

Angular rate ~constant within the update interval

$$\approx \mathbf{C}^{\alpha}_{\beta}(t)\exp\left(\mathbf{\Omega}^{\beta}_{\alpha\beta}\delta t\right)$$

Series expansion of exp

$$= \mathbf{C}^{\alpha}_{\beta}(t)\sum_{r=0}^{\infty}\frac{[\mathbf{\Omega}^{\beta}_{\alpha\beta}\delta t]^r}{r!}$$

First order approximation (small angles & dt)

$$\approx \mathbf{C}^{\alpha}_{\beta}(t)(\mathbb{1}+\mathbf{\Omega}^{\beta}_{\alpha\beta}\delta t)$$

$$\mathbf{C}^{\alpha}_{\beta}(+) = \mathbf{C}^{\alpha}_{\beta}(-)(\mathbb{1}+\mathbf{\Omega}^{\beta}_{\alpha\beta}\delta t) \Leftarrow \textbf{3D}$$

$$\Psi(+) = \Psi(-)+\omega^{b}_{eb}\cdot\delta t \Leftarrow \textbf{1D}$$

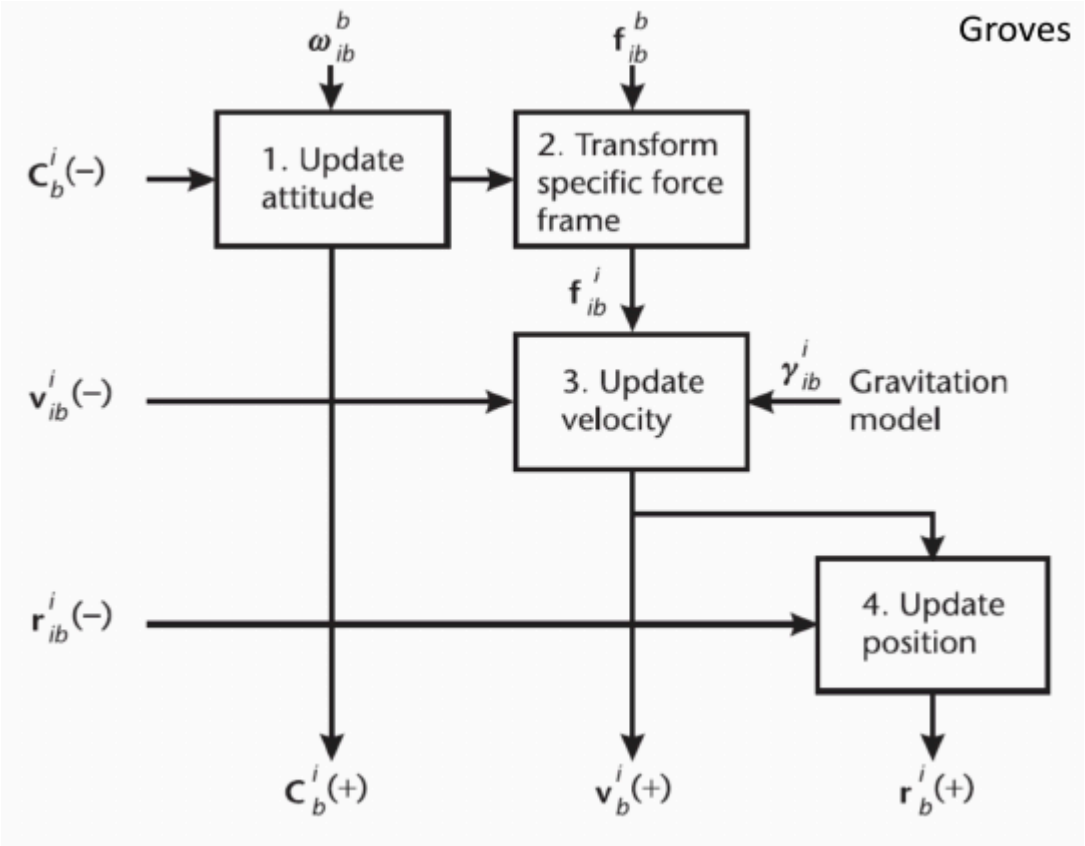
3D Inertial Navigation

- Wanted: position, velocity and attitude with respect to and resolved in the **inertial frame**
- Simple, because inertial sensors measure with respect to the inertial frame
- Only two frames are necessary, Earth rotation does not need to be considered

1. **Attitude update** (based on Gyroscope)
   - Integrate gyroscope readings to update the rotation matrix
   - Rotation matrix is defined between body and inertial frame (system of interest)
   - Gyroscopes measure angular rates between body and inertial frame
   - $\boldsymbol{C}^{i}_{b}(+) = \boldsymbol{C}^{i}_{b}(1_3+\mathbf{\Omega}^{b}_{ib}\delta t)$
   - Approximated angular rates: $\mathbf{\Omega}^{b}_{ib} = \begin{bmatrix} 0 & -\omega^{b}_{ib,z} & \omega^{b}_{ib,y} \\ \omega^{b}_{ib,z} & 0 & \omega^{b}_{ib,x} \\ -\omega^{b}_{ib,y} & \omega^{b}_{ib,x} & 0 \end{bmatrix}$
2. **Transformation** of specific-force resolving axis (body frame) to the **coordinate frame** used for position and velocity (usually the e-frame)
   - Accelerometer measurements $\boldsymbol{f}^{b}_{ib}$ (specific force): b-frame -> inertial frame
   - Rotation matrix:
   - $\boldsymbol{f}^{i}_{ib}(t) = \boldsymbol{C}^{i}_{b}(+)\boldsymbol{f}^{b}_{ib}$
3. **Velocity Update**, including specific **force** AND **gravitational acceleration** (e.g. using a gravity model)

- Assuming constant acceleration during update interval (integration of acceleration)
  - $\boldsymbol{v}_{ib}^{i}(+) = \boldsymbol{v}_{ib}^{i}(-) + \boldsymbol{a}_{ib}^{i}\delta t$
- Acceleration consists of specific force (measured by sensor) And gravity
  - $\boldsymbol{a}_{ib}^{i} = \boldsymbol{f}_{ib}^{i} + \gamma_{ib}^{i}$
    - γ_{ib}^{i} from gravity model
    - Or measured locally with known **rotation** and stationary sensor

4. **Position update**
   - Integrate acceleration
   - Assuming velocity is linear function of time during interval
   - $\boldsymbol{r}_{ib}^{i}(+) = r_{ib}^{i}(-) + v_{ib}^{i}(+)\delta t$



5. Optional. Transform attitude, velocity, position to useful frame
   - Use fancy super precise higher order integration schemes, if needed
   - Use implementation in other frames (Earth or local navigation frame), if needed
     - I to E-frame: Include Centrifugal Coriolis and Euler force in step 2 and additional rotation

Error Propagation

Ins Error Sources

Sensor readings ↓ True value ↓

$$\boldsymbol{\omega}_m = \mathbf{MK}\boldsymbol{\omega}_t + \mathbf{b}_g + \epsilon_g$$

$$\mathbf{M} = \begin{pmatrix} 1 & -\alpha_{yz} & \alpha_{zy} \\ 0 & 1 & -\alpha_{zx} \\ 0 & 0 & 1 \end{pmatrix}$$

axis misalignment
usually compensated by calibration

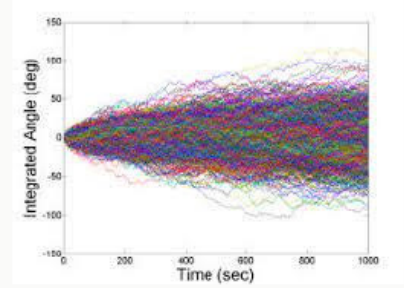
$$\mathbf{K} = \begin{pmatrix} k_{xx} & 0 & 0 \\ 0 & k_{yy} & 0 \\ 0 & 0 & k_{zz} \end{pmatrix}$$

scale factors
might change over time / temperature

$\mathbf{b}_g$ **bias**
change over time / temperature

ϵ_g **(white) noise**

Temperature effects are usually compensated by calibration for high grade sensors

- **Attitude Error**
  - $\theta_{err}(t) = \underbrace{b_{g0}t}_{\text{Start Bias}} + \underbrace{\int_0^t b_g'(\tau)d\tau}_{\text{Bias Instability}} + \underbrace{\int_0^t \epsilon_g(\tau)d\tau}_{\text{Angle Random Walk}}$
    - White Noise sequence/random walk with T as Sampling Interval
    - 

      Often given in datasheets:

      $$ARW = \sigma_\theta(t=1) = \sigma_g\sqrt{T} \quad \left[°/\sqrt{hr}\right]$$

      $$\sigma_\theta(t) = ARW \cdot \sqrt{t}$$
    - Alternatively: Noise Density : $ND[(°/s)/\sqrt{Hz}]$
      - $\sigma_g = ND \cdot \sqrt{1/T} = ND/\sqrt{T} = ARW/\sqrt{T}$
      - PSD - Power Spectral Density: $PSD = ND^2$
- Bias has constant and changing part
  - Constant part can be different every time the sensor is switched on (**Turn-On Bias Stability**)
  - Changing Part: **Bias (In-)Stability** in datasheets in °/hr
    - Usually std over a time span of 100s
- Bias drift as random walk (not 100% true)

Gyroscope Errors

$$\boldsymbol{\omega}_m = \mathbf{M}\mathbf{K}\boldsymbol{\omega}_t + \mathbf{b}_g + \epsilon_g$$

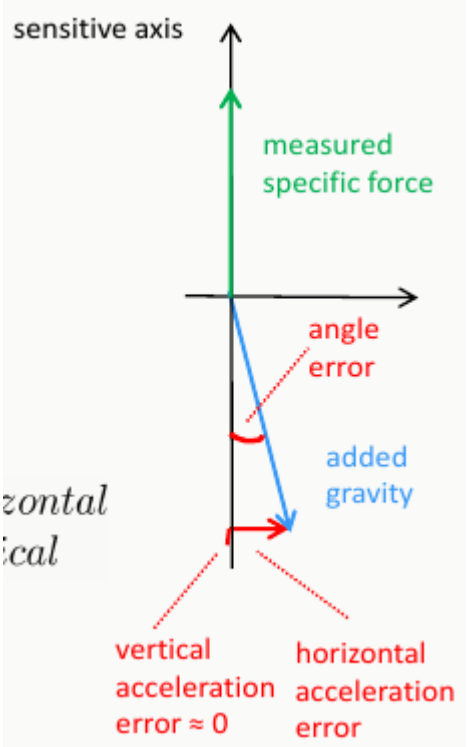
RLG	FOG	MEMS
$0.001 - 10\,{}^{\circ}/\mathrm{h}$	$0.1 - 50\,{}^{\circ}/\mathrm{h}$	$5\,{}^{\circ}/\mathrm{h} - 5\,{}^{\circ}/\mathrm{s}$
$5\,\mathrm{ppm}$	$> 100\,\mathrm{ppm}$	$> 400\,\mathrm{ppm}$
$0.01\,{}^{\circ}/\sqrt{\mathrm{h}}$	$0.1\,{}^{\circ}/\sqrt{\mathrm{h}}$	$1\,{}^{\circ}/\sqrt{\mathrm{h}}$

Accelerometer errors

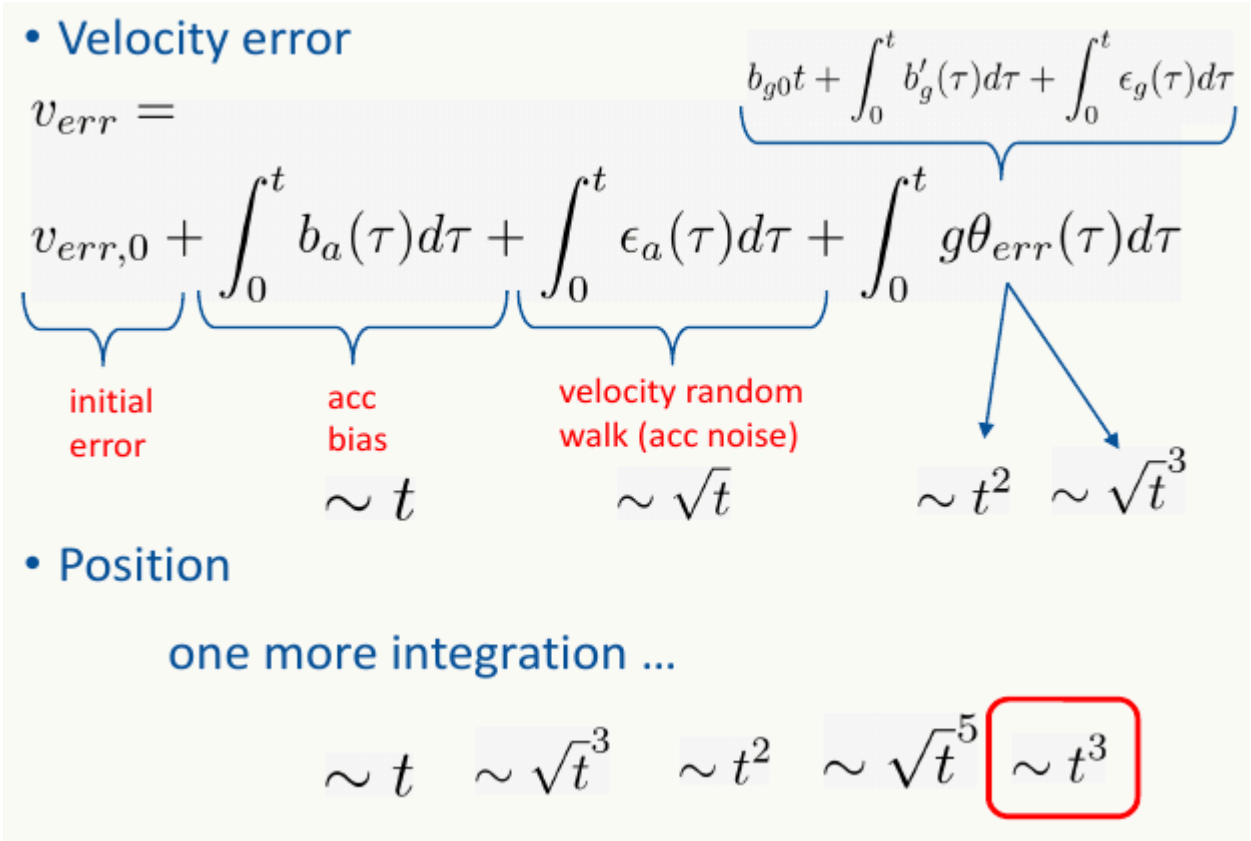
$$\mathbf{f}_m = \mathbf{M}\mathbf{K}\mathbf{f}_t + \mathbf{b}_a + \epsilon_a$$

	Vibrating Beam	Pendel
constant	$0.1 - 1\,\mathrm{mg}$	$0.1 - 10\,\mathrm{mg}$
changing	$0.1\,\mathrm{mg}$	$1\,\mathrm{mg}$
	$100\,\mathrm{ppm}$	$1000\,\mathrm{ppm}$
Velocity RW	$0.01\,{}^{\mathrm{m}}/_{\mathrm{s}}/\sqrt{h}$	$0.04\,{}^{\mathrm{m}}/_{\mathrm{s}}/\sqrt{h}$

- $\boldsymbol{a}_{ib}^n = \boldsymbol{C}_b^n \boldsymbol{f}_{ib}^b + \gamma^n$
  - accelerometer measurements
  - single axis
  - assume no scale factor error
  - angle error leads to vertical acceleration offset



Position and Velocity Error

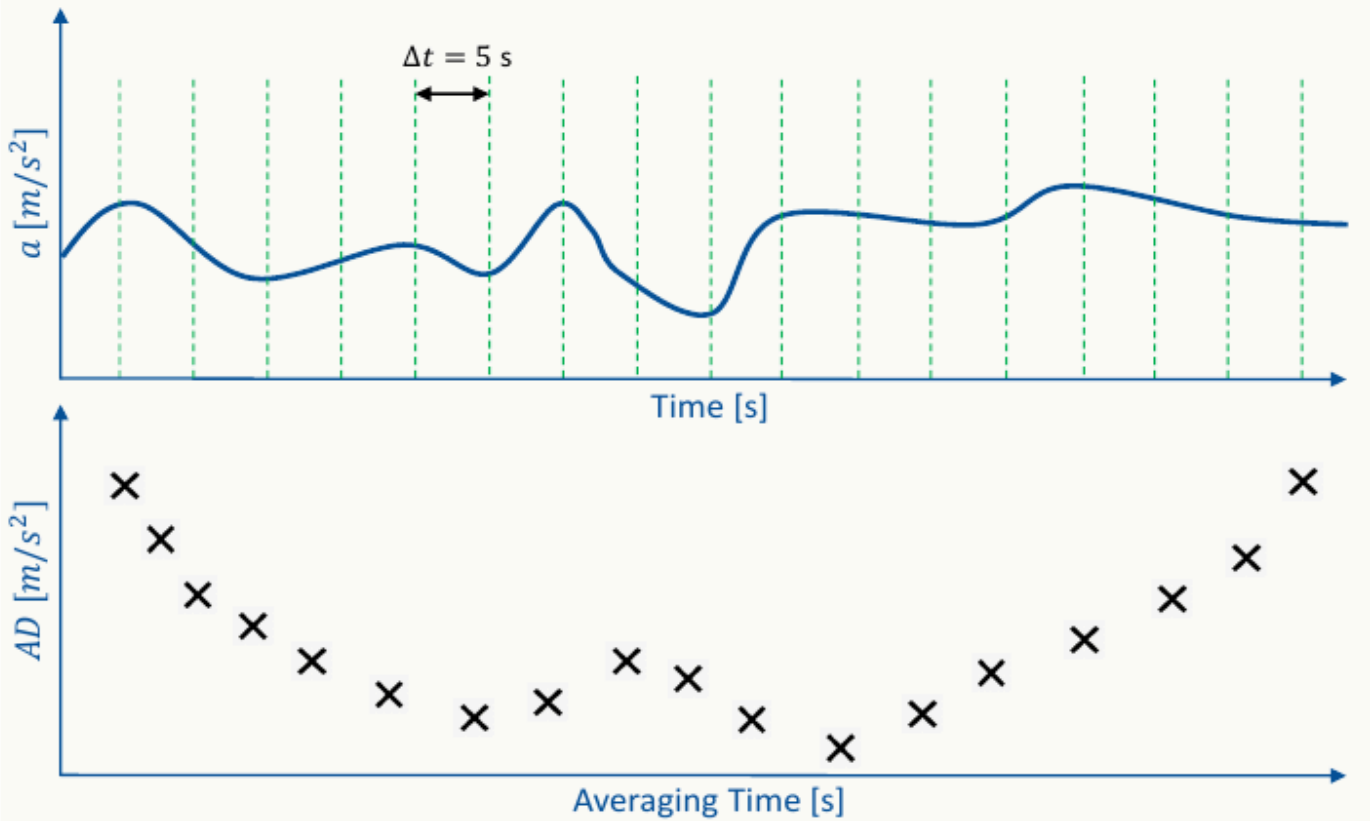


Allan Variance

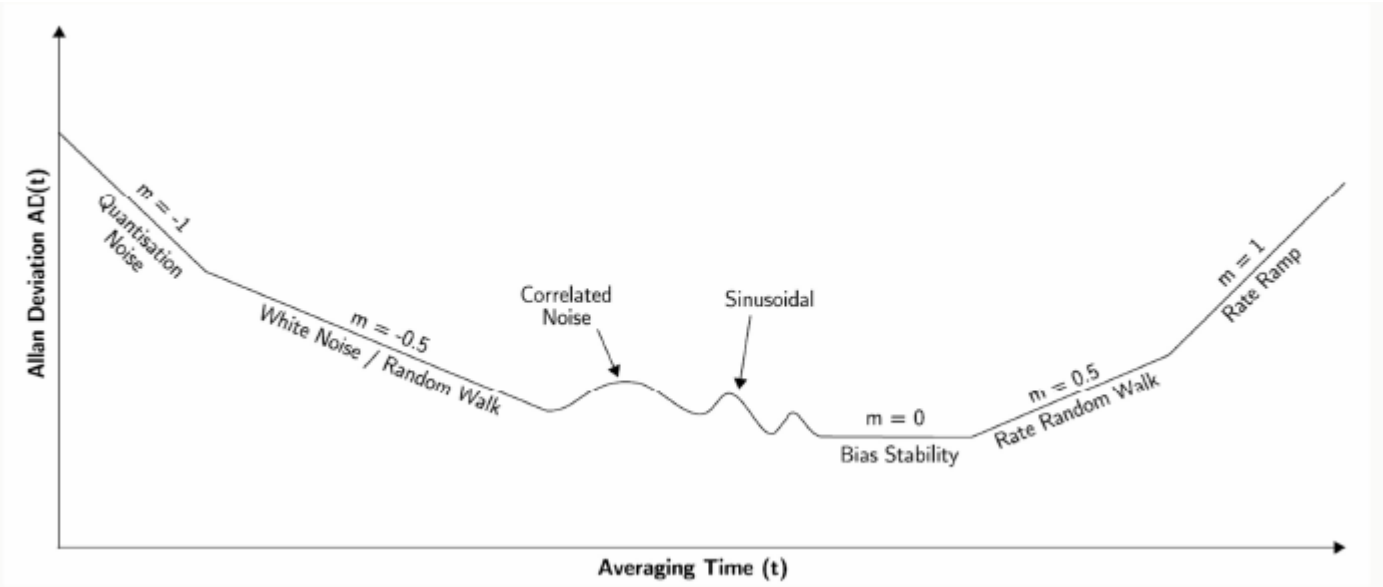
- Originally developed to analyse the frequency stability of clocks and oscillators
- Useful to quantify noise processes in inertial sensors
- Calculation
  - Build intervals of different length

- $[a(\Delta t)_1, a(\Delta t)_2, ...a(\Delta t)_n]$

- Calculate variance of consecutive interval mean differences
  - $\sigma_{al}^2(\Delta t) = \frac{1}{2\cdot(n-1)} \sum_i (a(\Delta t)_{t+1} - a(\Delta t)_i)^2$

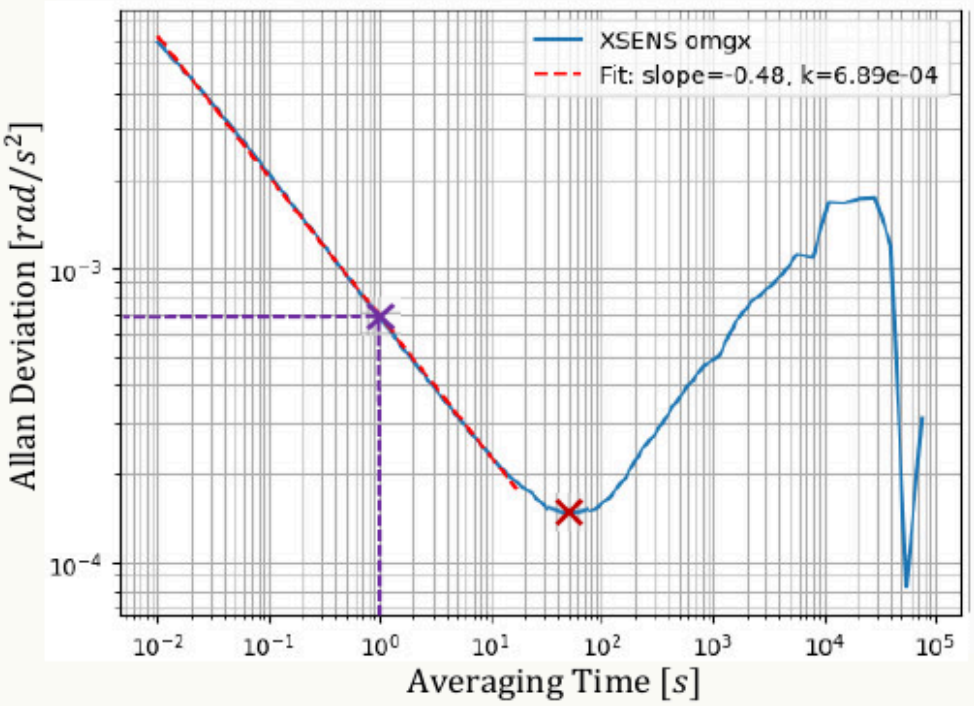


Allan Deviation (Idealised)



DREHRATE

Xsens



Angular Random Walk (VRW):

$$ARW = 6.9 \cdot 10^{-4} \frac{rad}{s} / \sqrt{s}$$

Bias Instability:

$$B_i = 1.4 \cdot 10^{-4} \frac{rad}{s} / \sqrt{s}$$

→ Exercise